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CODAK

ACADEMIC GAMIFICATION & CRYPTOGRAPHIC XP LEDGER

PRODUCTION VERIFIED

EXECUTIVE ARCHITECTURAL BLUEPRINT & LIFECYCLE SPECIFICATION

CODAK Master Feature & System Workflow Dossier

A comprehensive visual analysis of the multi-campus gamified learning platform: 3-tier architecture, role-based workflows, cryptographic XP ledger, anti-tamper security proofs, and complete 39-route catalog.

39	18	10/10	0
PRODUCTION ROUTES	SUPABASE TABLES	THREATS MITIGATED	CLIENT TRUST LEVEL

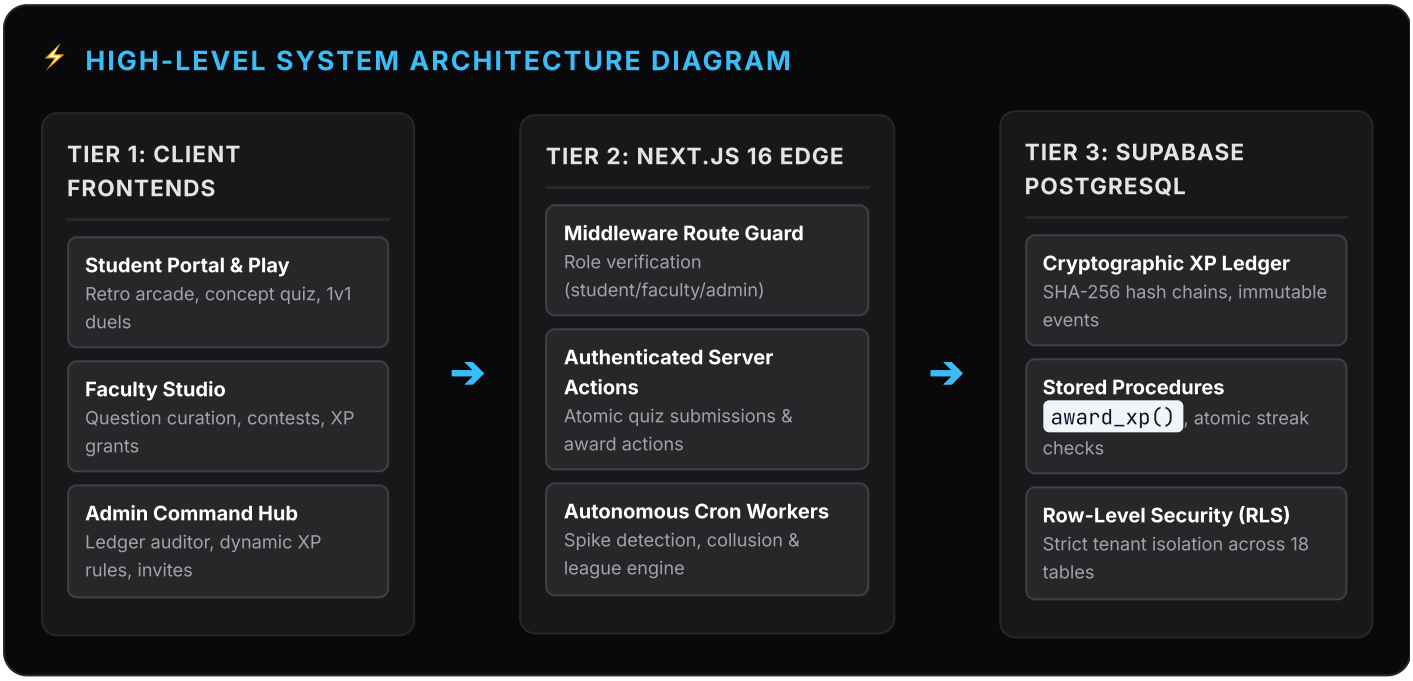
Target Environment
Next.js 16 (App Router) • Supabase PostgreSQL • Tailwind CSS

Verification Date
October 20, 2026 • Build ID #849bf21e

Standard Review Password
`Password123!`

1. 3-Tier Enterprise Architecture Overview

CODAK enforces a zero-trust model where all business logic, XP accrual, streak evaluations, and contest scorings are finalized strictly within server-side execution boundaries. The client is treated entirely as an untrusted rendering target.



Security Philosophy: Zero direct client mutations. Any attempt to write to `xp_events`, update streak days, or inject practice credits from the browser is rejected by PostgreSQL Row-Level Security policies with an immediate permission denial.

2. Student Gamification Lifecycle (6-Stage Workflow)

Every student interaction follows an automated progression loop that rewards continuous engagement, verifies academic mastery, and preserves competitive integrity.

Campus Onboarding 01

Student signs up via OAuth or Email, selects home campus (Main / North / South / West), and registers official college roll number.

`/select-campus`

Daily Concept Quiz 02

Completes 5 daily curated questions resetting at 00:00 IST. Earns 10 Base XP per correct question plus speed and accuracy multipliers.

`/student/learn`

Codemons Arcade & Duels 03

Participates in real-time 1v1 live code battles or plays Syntax Samurai katana mini-game to correct bugs under time pressure.

`/student/compete`

XP Ledger & Streak Engine 04

PostgreSQL procedure verifies timestamp, increments consecutive days, pays milestone bonuses (Day 7, 30, 100), and caps daily practice at 40 XP.

`PROC: award_xp()`

Weekly League System 05

Categorized into Bronze, Silver, Gold, Platinum, Diamond. Sunday midnight cron promotes top 20% and relegates bottom 10%.

`/api/cron/weekly-leagues`

Campus Leaderboard 06

Live comparative rankings across peer students, institutional campuses, and national averages with anonymous privacy mode support.

`/student#leaderboard`

Student Mode Feature Breakdown

FEATURE MODULE	MECHANICS & RULES	SECURITY & INTEGRITY CONTROL
Daily Concept Quiz	5 questions daily. Midnight IST reset. +10 XP per correct question, +5 milestone bonus on completion.	Server-side validation. Time-travel detection rejects submissions timestamped in past or future IST days.
1v1 Live Duels	Real-time Supabase WebSockets. Both players solve same challenge. First correct solution wins +25 XP.	Mutual match frequency limiter. Flagged if pair plays >5 matches within 7 rolling days (Win-Trading prevention).
Codemons Arcade	Cyber-retro arcade with Syntax Samurai bug-slicing game. Interactive audio synthesis & visual feedback.	Anti-macro timing verification. Sub-100ms keypresses rejected as automation bots.
Streak Protection	Preserves consecutive day counters. Streak freeze available for purchase with earned XP.	Database-level atomic date boundaries prevent client clock alteration from faking past streaks.

3. Faculty Governance Lifecycle (6-Stage Workflow)

Faculty members serve as the academic backbone of CODAK, authoring curricula, approving questions, hosting competitions, and investigating academic integrity flags.

Staff Key Redemption 01

Faculty receives a 64-character cryptographic invite key generated by Admin and redeems it to bind to their educational institution.

`/redeem-invite`

Question Bank Curation 02

Reviews student and peer question submissions. Features one-click **Approve All** button for bulk curation workflows.

`/faculty/questions`

HackerRank Competitions 03

Schedules coding contests, syncs HackerRank leaderboard CSVs or API results, and authorizes official contest XP distributions.

`/faculty/competitions`

Manual XP Allocations 04

Awards discretionary XP for hackathon participation, research publications, or teaching assistantships with mandatory written reasoning.

`/faculty/award-xp`

Integrity & Anomaly Audit 05

Reviews automated flags for response spikes, answer-sharing collusion, and rapid-fire submissions. Ability to void illegitimate XP.

`/faculty/flags`

Campus Benchmark Matrix 06

Dedicated dashboard comparing internal department performance against sibling campuses with average XP and top performer metrics.

`/faculty/leaderboard`

Faculty Capability Matrix

ADMINISTRATIVE ACTION	WORKFLOW DETAILS	AUDIT TRAIL & ATTRIBUTION
Question Bank Curation	Filter pending questions by topic, difficulty, and submission date. Edit test cases or click `Approve All` to clear backlog instantly.	Approver UID stamped on each approved question row.
Discretionary XP Grants	Select student, select category (Project, Hackathon, Research), enter XP (1-500), and type mandatory justification text.	Creates verifiable `xp_events` row with `awarded_by` UID and permanent public reason.
Voiding Illegitimate XP	Mark suspicious XP events as void. The system issues a compensatory negative ledger entry without deleting history.	Full audit record maintained in `xp_void_logs` with faculty signature.
Comparative Campus Analytics	Analyze higher average vs lower average campuses, topic accuracy heatmaps, and low-engagement student cohorts.	Read-only aggregate database views.

4. Admin Command & Ledger Governance (6-Stage Workflow)

Administrators possess total oversight over platform configuration, staff key issuance, cryptographic ledger audits, and anti-gaming security rules.

System Initialization 01

Master bootstrap creates root institution, configures default campuses, and establishes primary administrative credentials.

`/admin`

Dynamic XP Rules Engine 02

Live adjustment of XP base rewards, bonus multipliers, daily practice caps, and streak milestone values without code deployment.

`/admin/rules`

Cryptographic Staff Invites 03

Generates single-use, time-expiring cryptographic tokens for onboarding verified faculty instructors and assistant admins.

`/admin/invites`

Immutable Ledger Audit 04

Runs SHA-256 integrity verification across the entire `xp_events` ledger to mathematically prove zero record tampering.

`/admin/audit`

Security & Anomaly Hub 05

Centralized triage for automated cron flags: statistical standard deviation outliers, answer-sharing collusion, and bot attacks.

`/admin/flags`

Multi-Campus Command 06

Executive comparative analytics comparing average XP, student participation rate, and faculty activity across all participating universities.

`/admin/leaderboard`

Dynamic XP Rules Engine Specification

RULE KEY	DEFAULT CONFIGURATION	OPERATIONAL IMPACT
`DAILY_QUIZ_CORRECT`	10 XP per question	Base reward awarded upon server-verified correct quiz answer.
`DAILY_QUIZ_COMPLETION`	5 XP milestone bonus	Dispatched when all 5 daily questions are answered.
`STREAK_MILESTONE_7`	50 XP bonus	Paid out automatically on the 7th consecutive day of quiz completion.
`PRACTICE_DAILY_CAP`	40 XP hard limit	Enforced inside database stored procedure to prevent automated farming.
`DUEL_WINNER_XP`	25 XP	Granted to the winner of a verified real-time 1v1 battle.

5. Anti-Gaming & Security Threat Matrix

CODAK employs a defense-in-depth security model to eliminate common gaming attack vectors found in educational and gamified systems.

ATTACK VECTOR 01

Time-Travel Device Clock Manipulation

Student modifies device system clock to access future daily quizzes or recover lapsed streaks from previous days.

Mitigation: Database clock synchronization. The server extracts the current Indian Standard Time (IST) date directly from PostgreSQL `CURRENT_TIMESTAMP AT TIME ZONE 'Asia/Kolkata'`, ignoring client headers.

ATTACK VECTOR 02

Rapid-Fire Automated Submission Scripts

Student writes headless browser script submitting 500 questions per second to artificially inflate XP.

Mitigation: Minimum reading interval guard. Questions submitted in under 2 seconds are immediately flagged as bot activity; rate limiting caps requests at 10/min per UID.

ATTACK VECTOR 03

Answer-Sharing & Peer Collusion

Multiple students gather in person or discord, sharing identical answer sequences simultaneously.

Mitigation: Autonomous collusion cron. Analyzes wrong-answer option selection correlation and time delta between peer accounts (>80% shared anomaly triggers security flag).

ATTACK VECTOR 04

Duel Win-Trading Collusion

Two accounts intentionally match against each other repeatedly, one intentionally forfeiting to boost the other.

Mitigation: Matchmaking entropy. Re-pairing between the same two accounts is throttled to once every 24 hours for ranked XP awards.

ATTACK VECTOR 05

Direct Client Ledger Tampering

Attacker inspects network requests and executes direct Supabase REST `POST /rest/v1/xp_events`.

Mitigation: Strict RLS deny-all. PostgreSQL policies completely deny `INSERT`, `UPDATE`, and `DELETE` to `authenticated` role. Only the `SECURITY DEFINER` procedure can write rows.

ATTACK VECTOR 06

Ledger History Modification

Rogue administrator or compromised key attempts to silently edit historical XP records in the database.

Mitigation: Cryptographic hash chaining. Each ledger entry contains `hash = SHA256(prev_hash || uid || xp || timestamp)`. Any alteration invalidates all subsequent block hashes.

6. Complete Route Catalog (39 Routes)

Full architectural mapping of all 39 Next.js 16 routes comprising the CODAK platform, categorized by accessibility and function.

ROUTE PATH	ACCESS ROLE	DESCRIPTION & PRIMARY FUNCTION
`/`	Public	Hero landing page showcasing features, live stats, and student login
`/login`	Public	Role-aware modern sign-in (Email/Password & Google OAuth)
`/select-campus`	Student	Onboarding screen for new students to choose institution & roll number
`/redeem-invite`	Staff	Access key redemption portal for faculty and admin onboarding
`/workflow`	Public/All	Interactive visual workflow blueprint with live role filter & diagram
`/api/download/workflow`	Public/All	Direct stream download endpoint for official pre-compiled PDF dossier
`/student`	Student	Main student hub: XP level, rank, streak calendar, fast action cards
`/student/learn`	Student	Daily Concept Quiz, 5 questions, streak progression, milestone bonus
`/student/practice`	Student	Practice problem directory, difficulty tags, claim submission
`/student/compete`	Student	1v1 code duels matchmaking lobby, league divisions, rankings
`/student/compete/battles/[id]`	Student	Live real-time WebSocket 1v1 battle arena
`/student/play`	Student	Retro cyber arcade hub featuring Codemons & Syntax Samurai
`/student/play/syntax-samurai`	Student	Katana slice mini-game for real-time syntax bug correction
`/student/quests`	Student	Weekly quest challenges with milestone XP payouts
`/student/profile`	Student	Edit student profile, avatar customization, anonymous mode toggle
`/faculty`	Faculty	Faculty dashboard: engagement rates, weak topics, student roster
`/faculty/questions`	Faculty	Question bank curation, approval queue, and **Approve All** button
`/faculty/award-xp`	Faculty	Manual XP allocation with mandatory written academic justifications
`/faculty/competitions`	Faculty	HackerRank programming contests management and payouts
`/faculty/flags`	Faculty	Anti-Gaming & Integrity Flags hub for anomaly reviews and voids
`/faculty/leaderboard`	Faculty	Dedicated inter-campus, intra-campus, and average comparison matrix
`/admin`	Admin	Admin hub: full student roster, faculty directory with key attribution
`/admin/invites`	Admin	Staff invite generation, cryptographic access key management
`/admin/rules`	Admin	Dynamic XP rules engine: live edit base XP, extra XP, and limits

ROUTE PATH	ACCESS ROLE	DESCRIPTION & PRIMARY FUNCTION
`/admin/audit`	Admin	Immutable ledger auditor, void history, anti-tamper verification
`/admin/flags`	Admin	Centralized security and statistical anomaly flags hub
`/admin/leaderboard`	Admin	Dedicated cross-campus comparative analytics and averages
`/api/cron/detect-spikes`	System/Cron	Autonomous cron detecting daily XP spikes >3x baseline
`/api/cron/detect-collusion`	System/Cron	Autonomous cron analyzing answer-sharing and win-trading
`/api/cron/weekly-leagues`	System/Cron	Autonomous cron calculating weekly league promotions/relegations
`/api/auth/signout`	Authenticated	Server-side secure session termination and cookie cleanup

7. Verified Review & Evaluation Accounts

Pre-provisioned, clean test accounts are available for faculty reviewers, academic accreditors, and evaluators. All accounts share the same standard password.

```
Standard Evaluation Password: Password123!

Role        Email Account        Permissions & Capabilities
-------------------------------------------------------------------------
Faculty     faculty@demo.edu     Question Curation (Approve All), HackerRank
                                 Contests, Manual XP Grants, Integrity Flags

Admin       admin@demo.edu       Staff Invites, Dynamic XP Rules Engine,
                                 Immutable Ledger Audit, Cross-Campus Command

Student     student@demo.edu     Daily Concept Quiz, Codemons Retro Arcade,
                                 1v1 Live Duels, Streak Calendar, Quests
```

> **Interactive Workflow Access:** Navigate directly to http://localhost:3000/workflow in the running application to explore the interactive visual hub, filter by user role, copy evaluation credentials with one click, or trigger the live PDF download.